

Breaking Enigma: Secrets, War, and the Birth of Computing

Group 9's Members:

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Group Topic Theme: **Historical computing**

As tensions escalated into war, so too did interest in cryptography, as nations sought for secure methods of communications. In 1913, the outbreak of World War I gave rise to cryptology in military communications as such the first cryptography rotor machine, the Hebern Rotor Machine, was made. Following the aftermath of the war, German cryptologist Arthur Scherbius developed the Enigma Machine, utilizing rotor circuits to both encode and decode, gaining influence before and during WWII. The proposed topic, *Breaking Enigma: Secrets, War, and the Birth of Computing*, aims to explore the history, design, and operation of the Enigma Machine, including its impact on cryptography, wartime intelligence, and early development of computing technologies.

The proposed topic will include the following interactive elements:

- **User Simulation** - The users will be able to simulate the operation of the enigma machine by entering a message and observing how it is encrypted using a simplified rotor-based cipher system. The simulation will visually demonstrate how each keystroke passes through the machine's components. Furthermore, the users will be able to modify rotor positions, observing how minor configurations will produce vastly different encrypted outputs.
- **Interactive Historical Timeline** - Users can explore key events related to the development of the Enigma Machine, its use before and during World War II, and the codebreaking efforts that contributed to the emergence of early computing technologies. They may select individual events to view additional information, historical images, diagrams, and notable figures involved.
- **Codebreaking Challenge** - Users will be presented with encrypted messages and clues, allowing them to experience the challenges faced by wartime cryptanalysts and gain a better understanding of cryptographic principles. Using the knowledge gained, the users must attempt to decipher the messages within a limited number of attempts, gradually increasing in difficulty, introducing different concepts the higher the difficulty.

Guide Snapshot

Project Info

- **Title:** Breaking Enigma: Secrets, War, and the Birth of Computing
- **Theme:** Historical computing
- **Members:** (Dustin Josh Bandigan, Nathan James Catapang, Raeka Estrella Hernaez, Paul Rhazzel Serrano, Chelsea Nichole Tamayo)

Color Palette

Aiming for an old war type vibe, Monochrome/Muted colors

- **Background:** #F5F5F5 (White-ish Gray)
- **Primary Text/UI:** #1A1A1A (Near Black)
- **Accents & Buttons:** #8E8E8E (Medium Gray)
- **Subtle UI Elements:** #D4D4D4 (Light Gray)
- **Cipher Output Text:** #4A4A4A (Dark Gray)
- **Others:** Any Color within the Grayscale spectrum

Typography

- **Headings & Titles:** *Special Elite or Courier Prime or Any Alternative* (Typewriter look)
- **Body Text:** *Inter or Open Sans* (Modern readability)
- **Cipher/Encrypted Data:** *Fira Code or Monospace* (Perfect character alignment)

UI & Interactive Guidelines

- **User Simulation:** Mechanical keystroke animations with glowing paths tracing the signal through the simplified rotor-based cipher system.
- **Rotor Settings:** Coded to allow for modifying of rotor positions.
- **Timeline Layout:** Horizontal path with clickable circular nodes for exploring key events.
- **Timeline Pop-ups:** Clean modal cards displaying historical images, diagrams, and notable figures.
- **Codebreaking Challenge:** Clues styled as classified manila dossiers.
- **Codebreaking Inputs:** Terminal-style text fields for attempting to decipher the messages.
- **Difficulty Scaling:** Subtle visual static added to the background as cryptographic concepts increase in difficulty.